SOFTWARE REQUIREMENTS SPECIFICATION

Programming Education Platform for Secondary Students

Combined Requirements and User Scenarios

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Project	Programming Education Platform for Secondary Students

Prepared as a single, organized baseline document

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Requirement identifiers are retained from the existing SRS. User scenarios are arranged by actor and cross-functional purpose for easier review.

1. Introduction

This project is an online programming education platform for secondary-school students. It provides a structured environment in which students can find suitable programming courses and study content created by teachers. The platform combines course discovery, structured lessons, assessments, progress tracking, feedback, guardian monitoring, and administrative controls for course quality and complaints.

The platform also supports the operational needs of teachers and teaching assistants, including course administration, grading, student performance analytics, and the management of eligible earnings from paid courses.

2. Purpose

The purpose of the system is to help secondary-school students learn programming through structured courses provided by teachers. Students can discover courses, enroll in free courses or purchase paid courses, study lessons, complete quizzes and assignments, track progress, and receive grades and feedback. Teachers can create and manage courses, assign teaching assistants, assess student work, monitor performance, and review their balances. Guardians can monitor their authorized children's academic progress, while administrators manage complaints, review paid courses before publication, and enforce platform policies.

3. Scope

3.1 In Scope

- User registration and login.
- Course browsing, course details, and course structure.
- Free and paid courses, enrollment, and payment.
- Course modules, lessons, mandatory module quizzes, optional lesson quizzes, and lesson assignments.
- Student progress tracking, assignment submission, quiz attempts, grading, grades, and feedback.
- Course reviews and ratings.
- Teacher course management and teacher dashboards.
- Teaching assistant assignment, assigned-course access, and grading.
- Guardian monitoring of linked children.
- Student complaints and administrative complaint handling.
- Administrative approval of paid courses and teacher compliance acknowledgement.
- Refund periods, delayed teacher payout, and teacher balance.
- Course lifecycle management.

3.2 Course Structure

Course → Modules → Lessons → Quizzes / Assignments

A course contains one or more modules. Each module has a mandatory module-level quiz. Lesson-level quizzes and lesson assignments are optional, according to teacher decisions and the defined access rules.

4. Stakeholders

Stakeholder	Interest and responsibility
Secondary-school students	Discover and study programming courses; enroll or purchase; submit quizzes and assignments; view grades and progress; review courses; raise complaints.
Teachers	Create and manage courses and learning materials; create assessments; assign teaching assistants; grade work; monitor performance; view eligible earnings.

Stakeholder	Interest and responsibility
Teaching assistants	Support assigned courses by viewing students, monitoring performance, and grading authorized assignments and quizzes.
Guardians	Monitor the enrolled courses, progress, quiz grades, and assignment grades of linked and verified children.
Administrators	Review paid courses, manage complaints and review moderation, review refund requests, and enforce platform policies.
Payment provider	Support course payment and refund transactions and provide transaction status to the platform.

5. System Actors

5.1 Student

Create an account; browse courses and course details; enroll in courses; study lessons; take quizzes; submit assignments; view grades and progress; submit reviews and complaints.

5.2 Teacher

Create and manage courses, modules, lessons, video links, quizzes, and assignments; assign teaching assistants; view students and performance; grade work; analyze course data; view balance.

5.3 Teaching Assistant

View assigned courses and their students; access ungraded submissions and quizzes requiring grading; grade authorized work; view student grades and performance.

5.4 Guardian

View a linked child’s enrolled courses, progress, quiz grades, and assignment grades.

5.5 Administrator

Review complaints and reported reviews; communicate with teachers; manage complaint cases and refund requests; review paid courses; verify readiness and acknowledgements; enforce platform policies and course lifecycle rules.

5.6 Payment Provider (supporting actor)

Process or confirm payment and refund transactions and provide status information used by the platform.

6. Functional Requirements

The requirements below retain the identifiers used in the existing SRS. The user scenarios in Section 9 provide the corresponding actor flows and common exceptions.

6.1 Authentication and Account Management

ID	Requirement
FR-AUTH-01	The system shall allow users to log in using a username and password.
FR-AUTH-02	The system shall allow users without an account to create a new account.

ID	Requirement
FR-AUTH-03	During registration, the user shall select their role. The currently supported public roles are: Student Teacher
FR-AUTH-04	The system shall collect different information depending on the selected role.
FR-AUTH-05	The system shall authenticate users before allowing access to protected functionality.
FR-AUTH-06	Users shall only have access to functionality corresponding to their role.
FR-AUTH-07	Administrator account creation shall be controlled by the platform and shall not be treated as normal public registration.

6.2 Course Discovery

ID	Requirement
FR-COURSE-01	The system shall display available courses using course cards. A course card may contain: Course image. Course name. Teacher name. School grade. Term. Course price. Free/Paid status.
FR-COURSE-02	Students shall be able to browse courses without being forced to select grade or term before browsing.
FR-COURSE-03	The system shall provide a course details page.
FR-COURSE-04	The course details page shall display relevant information such as: Course name. Teacher. Grade. Term. Course description. Number of enrolled students. Modules. Course ratings. Course reviews. Course price/status.
FR-COURSE-05	Students shall be able to inspect the course structure before enrollment.
FR-COURSE-06	A course shall contain grade and term information when it represents a school curriculum.
FR-COURSE-07	A course may cover: The complete curriculum. A part of the curriculum. A specific topic or group of topics.

ID	Requirement
FR-COURSE-08	Teachers shall be free to organize modules according to their teaching methodology. Modules do not have to exactly match the official curriculum units.

6.3 Course Structure

ID	Requirement
FR-CONTENT-01	A course shall contain one or more modules.
FR-CONTENT-02	Each module shall contain a mandatory module-level quiz.
FR-CONTENT-03	Each lesson shall support an assignment. The teacher decides whether to provide an assignment for the lesson.
FR-CONTENT-04	The teacher may optionally add a quiz to an individual lesson.
FR-CONTENT-05	Students shall be able to view module names and the available lesson/assessment structure before enrollment.
FR-CONTENT-06	Teachers shall be able to create and manage modules.
FR-CONTENT-07	Teachers shall be able to create and manage lessons.
FR-CONTENT-08	Teachers shall be able to provide external lecture/video links. The platform does not need to host uploaded video files.
FR-CONTENT-09	Teachers shall be able to create assignments for lessons.
FR-CONTENT-10	Teachers shall be able to create mandatory module quizzes.
FR-CONTENT-11	Teachers shall be able to create optional lesson quizzes.

6.4 Course Enrollment

ID	Requirement
FR-ENROLL-01	A student shall be able to enroll in multiple courses.
FR-ENROLL-02	The system shall clearly indicate whether a course is: Free Paid
FR-ENROLL-03	When a student enrolls in a free course, the student shall receive immediate access to the course content.
FR-ENROLL-04	When a student successfully purchases a paid course, the student shall receive immediate access to the course content.
FR-ENROLL-05	The student's enrolled courses shall be available from the student's account.

6.5 Course Access Rules

ID	Requirement
FR-ACCESS-01	The student shall be able to start learning from any module. The student does not have to complete Module 1 before accessing Module 2.
FR-ACCESS-02	Lessons inside the selected module shall follow sequential access.
FR-ACCESS-03	The student shall not be able to open the next lesson until the content of the previous lesson has been completed.
FR-ACCESS-04	Watching a lesson shall not necessarily be required before accessing its quiz or assignment. Therefore, the student may access a quiz or assignment without watching the corresponding lecture when the system's access rules allow it.
FR-ACCESS-05	The student shall not be required to wait for an assignment to be graded before continuing to the next available learning step.

6.6 Student Progress

ID	Requirement
FR-PROGRESS-01	The system shall track the student's progress within each enrolled course.
FR-PROGRESS-02	Progress shall take into consideration assessment activity, including: Quizzes. Assignments.
FR-PROGRESS-03	Video/lesson viewing activity shall also be tracked where applicable.
FR-PROGRESS-04	The system shall store student assessment results.
FR-PROGRESS-05	Students shall be able to view their own progress.
FR-PROGRESS-06	Teachers shall be able to view student progress.
FR-PROGRESS-07	Teaching assistants shall be able to view student progress for courses they assist in.
FR-PROGRESS-08	Guardians shall be able to view their child's progress.
FR-PROGRESS-09	The exact mathematical formula used to calculate the overall progress percentage shall be defined before implementation.

6.7 Quizzes

ID	Requirement
FR-QUIZ-01	Each module shall contain a mandatory quiz.
FR-QUIZ-02	Teachers may optionally add quizzes to individual lessons.
FR-QUIZ-03	Students shall be able to attempt available quizzes.
FR-QUIZ-04	The system shall store quiz results.

ID	Requirement
FR-QUIZ-05	Teachers shall be able to view student quiz results.
FR-QUIZ-06	Teaching assistants shall be able to view and grade authorized quizzes.
FR-QUIZ-07	Quiz results shall contribute to the student's course performance.

6.8 Assignments

ID	Requirement
FR-ASSIGN-01	Teachers shall be able to add an assignment to a lesson.
FR-ASSIGN-02	Assignments are optional for the teacher.
FR-ASSIGN-03	Students shall be able to submit assignments.
FR-ASSIGN-04	After submitting an assignment, the student shall be allowed to continue to the next available step without waiting for grading.
FR-ASSIGN-05	Teachers shall be able to view submitted assignments.
FR-ASSIGN-06	Teaching assistants shall be able to view submitted assignments for their assigned courses.
FR-ASSIGN-07	Teachers and authorized teaching assistants shall be able to grade assignments.
FR-ASSIGN-08	The system shall store assignment grades.
FR-ASSIGN-09	Students shall be able to view their assignment grades after grading.

6.9 Reviews and Ratings

ID	Requirement
FR-REVIEW-01	The system shall support course ratings and written reviews.
FR-REVIEW-02	A student shall only be allowed to review a course after completing at least one module.
FR-REVIEW-03	Course ratings and reviews shall be displayed on the course details page.
FR-REVIEW-04	The exact review moderation policy shall be defined before implementation.

6.10 Teacher Course Management

ID	Requirement
FR-TEACH-01	A teacher shall be able to create a course.
FR-TEACH-02	A teacher shall be able to create modules.
FR-TEACH-03	A teacher shall be able to create lessons.
FR-TEACH-04	A teacher shall be able to add lecture/video links.

ID	Requirement
FR-TEACH-05	A teacher shall be able to create quizzes.
FR-TEACH-06	A teacher shall be able to create assignments.
FR-TEACH-07	A teacher shall be able to manage their courses.
FR-TEACH-08	A teacher shall be able to add teaching assistants to a course.

6.11 Teacher Dashboard

ID	Requirement
FR-DASH-01	The teacher dashboard shall display student/course analytics.
FR-DASH-02	Analytics shall be presented using charts where appropriate.
FR-DASH-03	The teacher shall be able to view all students enrolled in their courses.
FR-DASH-04	The teacher shall be able to select a specific student.
FR-DASH-05	For a selected student, the teacher shall be able to view: Quiz grades. Assignment grades. Course progress. Learning activity. Number of videos/lessons viewed where tracked.
FR-DASH-06	The teacher shall be able to view course-level student performance.
FR-DASH-07	The dashboard shall display the teacher's balance.
FR-DASH-08	The teacher balance shall represent money accumulated from eligible course purchases according to the platform's payout policy.

6.12 Teaching Assistant

ID	Requirement
FR-TA-01	A teacher shall be able to add a teaching assistant to a course.
FR-TA-02	The system shall grant the teaching assistant access to the assigned course.
FR-TA-03	The teaching assistant shall see all courses in which they are assigned.
FR-TA-04	The teaching assistant shall be able to open an assigned course.
FR-TA-05	The teaching assistant shall be able to view all students in the course.
FR-TA-06	The teaching assistant shall be able to view assignments that have not been graded.
FR-TA-07	The teaching assistant shall be able to view quizzes requiring grading.
FR-TA-08	The teaching assistant shall be able to grade assignments.

ID	Requirement
FR-TA-09	The teaching assistant shall be able to grade quizzes where applicable.
FR-TA-10	The teaching assistant shall be able to view student grades and course performance.
FR-TA-11	The teaching assistant shall not have unrestricted administrative permissions.

6.13 Guardian

ID	Requirement
FR-GUARD-01	The guardian shall be able to view all courses in which their child is enrolled.
FR-GUARD-02	The guardian shall be able to view the child's progress in each course.
FR-GUARD-03	The guardian shall be able to view the child's quiz grades.
FR-GUARD-04	The guardian shall be able to view the child's assignment grades.
FR-GUARD-05	The exact mechanism used to link a guardian account to a student's account shall be defined before implementation.

6.14 Complaints and Feedback

ID	Requirement
FR-COMPLAINT-01	The student shall be able to submit a complaint or feedback regarding a teacher/course.
FR-COMPLAINT-02	The administrator shall receive the complaint.
FR-COMPLAINT-03	The administrator shall review the complaint.
FR-COMPLAINT-04	The administrator shall be able to contact the relevant teacher.
FR-COMPLAINT-05	The administrator shall investigate and handle the complaint according to platform policies.
FR-COMPLAINT-06	The administrator shall record the complaint status and outcome.

6.15 Paid Courses and Approval

ID	Requirement
FR-PAID-01	Teachers shall be able to create paid courses.
FR-PAID-02	A paid course shall not become available for purchase before administrative approval.
FR-PAID-03	The administrator shall contact the teacher during the approval process.
FR-PAID-04	The administrator shall verify that the teacher is serious about providing the course.

ID	Requirement
FR-PAID-05	The administrator shall verify that at least the first module is prepared.
FR-PAID-06	The teacher shall receive an official acknowledgement/undertaking before approval.
FR-PAID-07	The acknowledgement shall state that the teacher is required to follow: The declared course content. Applicable company policies.
FR-PAID-08	The acknowledgement shall state that violating the agreed content/policies may result in a financial penalty according to company policy.
FR-PAID-09	After successful approval, the paid course shall become available for students to purchase.

6.16 Paid Course Refund and Teacher Payout

ID	Requirement
FR-PAY-01	Students shall receive immediate access after successfully purchasing a paid course.
FR-PAY-02	The teacher shall not immediately receive the corresponding funds in their available balance.
FR-PAY-03	For courses where content is released progressively, the payout shall remain on hold for 7 days after the student's purchase.
FR-PAY-04	The seven-day period represents the period during which the student has the right to request a refund.
FR-PAY-05	For fully recorded courses, the refund/payout hold period shall be 24 hours.
FR-PAY-06	After the applicable refund period passes, the corresponding amount becomes available to the teacher according to the platform's payout policy.
FR-PAY-07	The teacher's balance shall represent funds that have become available according to the payout rules.
FR-PAY-08	The exact payment provider and financial transaction mechanism shall be defined separately.

6.17 Course Lifecycle

ID	Requirement
FR-LIFECYCLE-01	A course shall have a lifecycle of one year.
FR-LIFECYCLE-02	After the one-year lifecycle ends, the course shall be deleted according to the platform's course lifecycle policy.
FR-LIFECYCLE-03	The exact starting point used to calculate the one-year period shall be finalized before implementation.

ID	Requirement
FR-LIFECYCLE-04	The system shall define how historical student grades and financial records are handled when a course reaches the end of its lifecycle.

6.18 Business Rules

ID	Business rule
BR-01	A student can enroll in multiple courses.
BR-02	Free courses provide immediate access after enrollment.
BR-03	Paid courses provide immediate access after successful payment.
BR-04	Students can start from any module.
BR-05	Lessons inside a module are sequential.
BR-06	The student does not have to watch a lesson before accessing its quiz or assignment when access is available.
BR-07	Every module has a mandatory quiz.
BR-08	Lesson-level quizzes are optional.
BR-09	Lesson assignments are optional for the teacher.
BR-10	The student does not have to wait for assignment grading before continuing.
BR-11	A student must complete at least one module before reviewing a course.
BR-12	Teaching assistants can grade assignments and quizzes for their assigned courses.
BR-13	Paid courses require administrative approval before publication.
BR-14	Teacher payout is delayed to protect the student's refund period.
BR-15	The payout hold is 7 days for progressively released courses.
BR-16	The payout hold is 24 hours for fully recorded courses.
BR-17	A course has a one-year lifecycle.

6.19 Core System Data

The system will manage information related to:

- User accounts, roles, student profiles, teacher profiles, and guardian accounts.
- Courses, modules, lessons, and lecture/video links.
- Quizzes, quiz attempts, quiz grades, assignments, assignment submissions, and assignment grades.
- Student progress, course enrollments, course reviews, and course ratings.
- Teaching assistants and teacher analytics.
- Teacher balances, payments, refunds, and payouts.
- Complaints, administrative actions, paid-course approval, and teacher acknowledgements.

7. Non-Functional Requirements

7.1 Security

ID	Requirement
NFR-SEC-01	Passwords shall never be stored as plaintext.
NFR-SEC-02	The system shall use secure authentication mechanisms.
NFR-SEC-03	Role-based authorization shall prevent unauthorized access.
NFR-SEC-04	Students shall only access their own private academic information.
NFR-SEC-05	Teachers shall only manage courses and students associated with their courses.
NFR-SEC-06	Teaching assistants shall only access courses to which they are assigned.
NFR-SEC-07	Guardian access shall be limited to authorized children.
NFR-SEC-08	Administrative functions shall only be accessible to authorized administrators.

7.2 Privacy

ID	Requirement
NFR-PRIV-01	Student academic information shall only be accessible to authorized users.
NFR-PRIV-02	Guardian information shall only expose data belonging to the linked child.
NFR-PRIV-03	Private student information shall not be publicly displayed on course pages.
NFR-PRIV-04	The system shall have defined policies for data retention and deletion.

7.3 Performance

ID	Requirement
NFR-PERF-01	The system should provide responsive navigation during normal usage.
NFR-PERF-02	Course browsing should remain responsive as the number of courses increases.
NFR-PERF-03	Teacher dashboards and analytics should load within an acceptable time.
NFR-PERF-04	Assignment and quiz submissions shall provide clear confirmation of successful submission.

7.4 Reliability

ID	Requirement
NFR-REL-01	The system shall minimize data loss.
NFR-REL-02	Quiz submissions shall not be silently lost.
NFR-REL-03	Assignment submissions shall not be silently lost.
NFR-REL-04	Grades shall be stored reliably.
NFR-REL-05	Financial transactions shall be traceable.
NFR-REL-06	Important administrative actions should be auditable.

7.5 Usability

ID	Requirement
NFR-UX-01	The user interface shall clearly distinguish between different user roles.
NFR-UX-02	Course cards shall provide important course information at a glance.
NFR-UX-03	Students shall easily understand: Current module. Current lesson. Available next lessons. Pending assignments. Available quizzes. Progress.
NFR-UX-04	Teacher dashboards shall clearly display important analytics.
NFR-UX-05	Teaching assistants shall easily identify ungraded assignments and quizzes.
NFR-UX-06	Error messages shall be clear and understandable to non-technical users.

7.6 Maintainability

ID	Requirement
NFR-MAINT-01	The system should be designed so future educational features can be added without major redesign.
NFR-MAINT-02	The system should support future programming tracks.
NFR-MAINT-03	The system should allow future integration of the live code editor.
NFR-MAINT-04	Business rules such as payout periods and course policies should be maintainable as platform policies evolve.

7.7 Scalability

ID	Requirement
NFR-SCALE-01	The system should support increasing numbers of: Students. Teachers. Courses. Teaching assistants. Assessments. Enrollments.
NFR-SCALE-02	The increase in users and course activity should not cause unacceptable degradation of normal system functionality.

7.8 Financial Integrity

ID	Requirement
NFR-FIN-01	Payment records shall be accurate.
NFR-FIN-02	Teacher balances shall be calculated from valid transactions.
NFR-FIN-03	Refund-related transactions shall be traceable.
NFR-FIN-04	Payout-hold periods shall be applied consistently.
NFR-FIN-05	The system shall prevent accidental duplication of financial transactions.

8. Use Case Diagram

Reserved area for the project use case diagram.

9. User Scenarios

Each scenario is presented with its actors, preconditions where provided, main flow, alternative flows, and postconditions or business rules where applicable. The use case inventory uses a consistent identifier set so the detailed scenarios can be referenced unambiguously.

9.1 Use Case Inventory

ID	Use case	Primary actor
UC-01	Register account	Student, Teacher, Guardian, Teaching Assistant
UC-02	Log in	All roles
UC-03	Browse and inspect courses	All roles
UC-04	Enroll in a free course	Student
UC-05	Purchase a paid course	Student
UC-06	Request a refund	Student
UC-07	Study an enrolled course	Student
UC-08	Take a quiz	Student
UC-09	Submit an assignment	Student
UC-10	View grades and progress	All roles
UC-11	Submit a course review	Student
UC-12	Submit a complaint	Student
UC-13	Create and publish a course	Teacher
UC-14	Create and manage assessments (quizzes and assignments)	Teacher, Teaching Assistant, Administrator
UC-15	Assign a teaching assistant	Teacher
UC-16	Grade student work	Teacher, Teaching Assistant
UC-17	Monitor student performance	Teacher, Teaching Assistant, Guardian, Student
UC-18	Monitor course performance	All roles
UC-19	View earnings and balance	Teacher, Administrator
UC-20	Approve or reject a course	Administrator
UC-21	Manage complaints	Teacher, Teaching Assistant, Administrator, depending on complaint
UC-22	Moderate course reviews	All roles (administrator performs removal decisions)
UC-23	Review refund requests	Administrator
UC-24	Monitor a child's progress	Guardian
UC-25	Access assigned courses and grading queue	Teaching Assistant; Teacher when no assistant is assigned
UC-26	Manage course lifecycle	Administrator

9.2 Shared and Account Scenarios

UC-01 — Register account

Actor: Student, Teacher, Guardian, Teaching assistant | **Priority:** High

Preconditions:

- The user does not already have an account using the same unique identifier.

Main flow

1. The user opens the registration page.
2. The user selects Student or Teacher or Teaching assistant or Guardian.
3. The system displays the appropriate registration form.
4. The user enters the required information.
5. The system validates the submitted data.
6. The system creates the account.
7. The system directs the user to the login page after account creation.

Alternative flows

- Required fields are missing → the system highlights the missing information.
- The username is already taken → the system asks the user to choose another one.
- The information is invalid → the system displays a clear validation message.

UC-02 — Log in

Actor: All roles

Main flow

1. The user opens the login page.
2. The user enters a username and password.
3. The system validates the credentials.
4. The system identifies the user's role.
5. The system opens the corresponding dashboard.

Alternative flows

- Incorrect credentials → the system displays an error.
- The user attempts to access another role's protected page → access is denied.
- The account is not authorized to access the requested functionality → the system redirects the user or displays an access-denied message.

Postcondition: The authenticated user can access only the functionality permitted by their role.

UC-03 — Browse and inspect courses

Actor: All roles

Main flow

1. The student opens the Courses page.
2. The system displays available course cards.
3. The student browses courses after selecting a grade or term first.
4. The student opens a course.
5. The system displays its teacher, grade, term, price, modules, ratings, and reviews.
6. The student examines the course structure.
7. The student decides whether to enroll or purchase the course.

Alternative flows

- No courses match the student's interests → an empty state is displayed.
- A course is pending approval, archived, or unavailable → it is not offered for enrollment or purchase.
- The course has incomplete information → the system displays only the available information.

Postcondition: The student understands what the course offers before enrolling.

UC-10 — View grades and progress

Actor: All roles

Main flow

1. The student opens the course dashboard.
2. The system displays overall course progress.

3. The student views completed lessons and available next lessons.
4. The student checks quiz results and assignment grades.
5. The system distinguishes graded assessments from pending ones.

Alternative flows

- If an assignment or written quiz has not been graded, the system displays its pending status rather than inventing a final grade.

Postcondition: The student can understand their current academic standing.

UC-17 — Monitor student performance

Actor: Teacher, Teaching Assistant, Guardian, Student

Main flow

1. The user opens the dashboard.
2. The system displays course-level analytics and charts.
3. The user selects a course.
4. The system displays its enrolled students and performance indicators.
5. The teacher or teaching assistant selects a student; the guardian selects their linked child; the student views their own analytics.
6. The system displays quiz grades, assignment grades, course progress, and tracked learning activity.
7. Each user reviews the student performance information permitted for their role.

Alternative flows

- The course has no enrolled students → the system displays an empty state.
- A student has no graded assessments → the system shows pending or unavailable results.
- Some learning activity is not tracked → the dashboard does not present data at all.

Postcondition: The teacher can monitor individual and overall course performance without accessing unrelated students' private information.

UC-18 — Monitor course performance

Actor: All roles

Main flow

1. The user opens the course performance analytics section
2. The system displays high-level metrics and trends for the course.
3. The user reviews course performance indicators relative to their role permissions.

Alternative flows

- Some learning activity is not tracked → the dashboard does not present data at all.

9.3 Student Scenarios

UC-04 — Enroll in a free course

Actor: Student | **Precondition:** Logged in

Main flow

1. The student opens an approved free course.
2. The student selects Enroll.
3. The system verifies that enrollment is allowed.
4. The system creates the enrollment record.
5. The system grants immediate access.
6. The course appears in My Courses.

Alternative flows

- The student is not logged in → the system requests authentication.
- The student is already enrolled → the system opens the existing enrollment instead of creating a duplicate.
- The course is unavailable for this student → enrollment is rejected.

Postcondition: The student can access the course content.

UC-05 — Purchase a paid course

Actor: Student | **Supporting actor:** Payment Provider

Main flow

1. The student opens an approved paid course by admin.
2. The student selects Purchase.
3. The system displays the price and payment instructions.
4. The student completes payment using the supported payment method.
5. The system verifies the payment result.
6. The system records the successful transaction and enrollment.
7. The student receives immediate access.
8. The system places the teacher's corresponding funds on hold for the applicable refund period.

Alternative flows

- Payment fails → no successful paid enrollment is created.
- Payment is pending → access is granted only after success is verified.
- Payment succeeds but the confirmation by the Payment Provider is delayed → the system checks the transaction status before creating another charge or duplicate enrollment.
- The student is already enrolled → the system prevents duplicate purchase or handles the existing enrollment according to policy.

Postcondition: Successful payment, enrollment, and teacher payout-hold records are consistent.

UC-06 — Request a refund

Actor: Student | **Reviewer:** Administrator

Main flow

1. The student opens the purchased course or transaction details.
2. The student submits a refund request and explains the reason.
3. The system records the request.
4. The administrator reviews the reason and relevant transaction information.
5. The administrator approves or rejects the request according to policy.
6. If approved, the refund is processed using the supported refund procedure.
7. The system records the outcome and updates the relevant financial records.

Alternative flows

- The request falls outside the applicable refund period → it is handled according to the final refund policy.
- The student reports a fraudulent teacher or an incorrect course → the administrator investigates.
- The request is rejected → the student can see the decision and reason.
- The refund transaction fails → the request remains unresolved until the issue is handled.

UC-07 — Study an enrolled course

Actor: Student

Main flow

1. The student opens My Courses.
2. The student selects an enrolled course.
3. The student chooses any module.
4. The system displays the lessons and assessments in that module.
5. The student opens the first available lesson.
6. The student completes the required lesson content.
7. The next lesson becomes available.
8. The student continues until the module's mandatory quiz is available.
9. The system records activity and updates progress.

Alternative flows

- The student attempts to open a locked next lesson → the system explains that the previous lesson must be completed.

- The student selects a different module → access can be permitted without completing earlier modules according to the teacher's policy.
- The student has not watched the lecture but wants to attempt an available assessment → the system denies.
- The student leaves before completing a lesson → the system preserves the recorded progress where supported.

Postcondition: Learning activity and assessment results are recorded.

UC-08 — Take a quiz

Actor: Student

Main flow

1. The student opens an available lesson quiz or mandatory module quiz.
2. The system checks assessment availability and the remaining attempt allowance.
3. The student answers the questions.
4. The student submits the quiz.
5. The system saves the attempt.
6. Automatically graded questions are graded.
7. Written questions, if any, enter the authorized grading queue.
8. The final result is stored and displayed when grading is complete.
9. The system updates the student's performance and progress.

Alternative flows

- The quiz is unavailable → the system prevents access.
- The student has exhausted the allowed attempts → another attempt is denied.
- Submission fails → the system provides a clear error and avoids silently losing the attempt.
- The quiz contains written questions → the result may remain pending until manual grading is complete.

UC-09 — Submit an assignment

Actor: Student

Main flow

1. The student opens an available lesson assignment.
2. The student reads the instructions.
3. The student submits the required work using the supported submission method.
4. The system validates and records the submission.
5. The system confirms successful submission.
6. The student continues to the next available learning step without waiting for grading.
7. The assignment appears in the assistant's grading queue.
8. After grading, the student can view the grade.

Alternative flows

- No assignment exists for the lesson (for example, an introductory lesson) → the student continues without submitting one.
- The submission is empty or invalid → the system requests a valid submission.
- The submission fails → the system informs the student and does not falsely mark it as submitted or falsely put it in the grading queue .
- The work has not been graded → the student sees a pending status.

UC-11 — Submit a course review

Actor: Student

Main flow

1. The student completes at least one module.
2. The system enables the review feature.
3. The student submits a rating or written review or both.
4. The system validates and stores the review.
5. The review appears on the course details page according to the moderation policy.

6. The student may edit their existing review and it will be marked as edited.

Alternative flows

- The review contains prohibited abuse or insults → an administrator may remove it under the moderation policy; the platform may warn the student, restrict future rating rights, or allow another opportunity to submit an acceptable review while preserving course access.
- A teacher disputes a review → the teacher submits a moderation request to the administrator.

Business rule: One review per student per course, editable by its author; spam and abusive content must be controlled.

UC-12 — Submit a complaint

Actor: Student

Main flow

1. The student opens the complaints or feedback section.
2. The student selects the relevant course or teacher.
3. The student describes the issue.
4. The system creates a complaint record.
5. The administrator receives the complaint.
6. The administrator investigates and contacts the teacher when necessary.
7. The complaint status and outcome are recorded.
8. Relevant parties are informed when required.

Alternative flows

- Required complaint details are missing → the system requests the missing information.
- The complaint requires teacher clarification → its status changes to Waiting for Teacher, if that status is implemented.
- The complaint is resolved or rejected → the final status and decision are recorded.

9.4 Teacher Scenarios

UC-13 — Create and publish a course

Actor: Teacher | **Supporting actor:** Administrator

Preconditions:

- The teacher is authenticated.
- The teacher has permission to create courses.

Main flow

1. The teacher opens the dashboard and selects Create Course.
2. The teacher enters the course title, description, image, grade (for example, first or second grade), term (first, second, or summer), and other applicable details.
3. The teacher selects whether the course is free or paid.
4. The teacher sets the price if it is a paid course.
5. The teacher creates one or more modules.
6. The teacher adds lessons and lecture/video links.
7. The teacher adds mandatory module quizzes.
8. The teacher optionally adds lesson quizzes and assignments.
9. The teacher saves the course and submits it for approval.
10. The administrator reviews the course.
11. After approval, the course becomes available to students according to its status and type.

Alternative flows

- Required course information is missing → the system requests completion.
- A module has no mandatory quiz → the system prevents publication until the requirement is satisfied.
- A paid course is missing its required first lesson → approval is withheld.
- The administrator rejects the course → the teacher receives the decision and can revise and resubmit it.
- The course is still pending approval → the course is hidden for all users except the admin and the teacher who posted it till it’s approved.

UC-14 — Create and manage assessments (quizzes and assignments)

Actor: Teacher, Teaching Assistant, Administrator

Main flow

1. The teacher opens a course they own.
2. The teacher selects a module or lesson.
3. The teacher creates a mandatory module quiz, a lesson quiz, or a lesson assignment.
4. For a quiz, the teacher defines the questions, answers, grading method, and attempt limit.
5. For an assignment, the teacher defines the instructions and submission requirements.
6. The teacher saves the assessment.
7. The system makes it available according to the course access rules.
8. Admins can delete the quiz or assignment if there is inappropriate content.

Alternative flows

- The teacher tries to remove the mandatory module quiz → the system prevents the module from violating the mandatory-quiz rule.
- The assessment contains incomplete questions or instructions → the system requests correction.
- The teacher changes an assessment after students have submitted work → the system preserves existing attempts and grades or applies the defined change policy; the system notifies students of the change and keeps the previous version available to students who already completed it.

Postcondition: Students can access the assessment when its availability conditions are met.

UC-15 — Assign a teaching assistant

Actor: Teacher

Main flow

1. The teacher opens a course they own.
2. The teacher selects Manage Teaching Assistants.
3. The teacher identifies the person to assign.
4. The system validates the account and assignment eligibility.
5. The system grants access to the selected course.
6. The assigned course appears in the teaching assistant's dashboard.

Alternative flows

- The specified account does not exist or is ineligible → the system displays an error.
- The person is already assigned → the system prevents duplicate assignments.
- The teacher attempts to assign someone to a course they do not own → the operation is denied.

UC-16 — Grade student work

Actor: Teacher, Teaching Assistant

Main flow

1. The authorized grader opens an assigned course.
2. The system displays ungraded assignments and assessments requiring manual grading.
3. The grader selects a submission.
4. The grader reviews the student's work.
5. The grader enters a grade and feedback; the system stores the feedback and notifies the student and guardian.
6. The system validates and stores the result.
7. The student can view the grade.
8. Student performance analytics are updated.

Alternative flows

- The work has already been graded → the system displays the existing grade and applies the permitted editing rules.
- The grade is outside the allowed range → the system requests a valid grade.
- A quiz contains only automatically graded MCQs → the system calculates those results without requiring manual grading.

Authorization rule: A teaching assistant can grade only within assigned courses and cannot access unrestricted administrative functions.

UC-19 — View earnings and balance

Actor: Teacher, Administrator | **Supporting actor:** Payment Provider

Main flow

1. The teacher opens the earnings section.
2. The system retrieves valid course purchase records.
3. The system separates funds on hold from eligible earnings.
4. The system calculates the teacher's available balance according to the payout policy.
5. The teacher reviews the balance and relevant transactions.

Alternative flows

- A purchase is still within its refund period → its funds remain on hold.
- A refund is approved → the relevant financial records are adjusted.
- A payment is pending or unsuccessful → it is not counted as an eligible completed purchase.

Rule: Progressively released courses: 7-day hold.

Rule: Fully recorded courses: 24-hour hold.

Rule: The teacher's share is initially 90% of the eligible amount, subject to the final fee and payout rules.

Scope note: This scenario covers viewing the teacher balance; teacher withdrawal or payout-request processing is outside this scenario.

9.5 Teaching Assistant Scenarios

UC-25 — Access assigned courses and grading queue

Actor: Teaching Assistant; the Teacher may perform these tasks when no teaching assistant is assigned.

Main flow

1. The assistant logs in.
2. The system displays the courses assigned by teachers.
3. The assistant selects a course.
4. The system displays enrolled students and course performance.
5. The assistant opens the ungraded assignments or quizzes requiring manual grading.
6. The assistant selects a submission and grades it.
7. The system saves the result and updates the student's records.

Alternative flows

- The assistant has no assigned courses → the dashboard displays an empty state.
- A course assignment is removed → the grading queue is updated to remove submissions associated with that assignment, according to policy.
- An assessment does not require manual grading → it does not appear in the manual grading queue.
- The assistant attempts to modify course ownership, payment records, or administrative settings → access is denied.

Postcondition: The assistant completes authorized academic tasks within the assigned courses.

9.6 Guardian Scenarios

UC-24 — Monitor a child's progress

Actor: Guardian

Preconditions:

- The guardian is authenticated.
- The guardian's account is linked to the correct students and the link has been verified.

Main flow

1. The guardian logs in.
2. The guardian opens the linked child's profile.

3. The system displays the child's enrolled courses.
4. The guardian selects a course.
5. The system displays the child's progress, quiz grades, and assignment grades.
6. The guardian reviews the available academic information.

Alternative flows

- No child is linked → the system requests account linking with the student.
- The phone number is not verified → the system does not establish the link.
- The guardian attempts to access another student's profile without verifying being his/ her guardian → access is denied.
- A grade is awaiting manual assessment → the system shows it as pending.

Postcondition: The guardian can monitor only the academic information of the authorized children.

Account-linking rule: A verified phone number is required; an email address is optional.

9.7 Administrator Scenarios

UC-20 — Approve or reject a course

Actor: Administrator

Main flow

1. The administrator opens the pending courses queue.
2. The administrator selects a course.
3. The system displays the course details, modules, lessons, assessments, and pricing information.
4. The administrator contacts the teacher to verify that he is ready.
5. The administrator verifies that at least the first lesson of a paid course is prepared.
6. The administrator provides the required compliance acknowledgement.
7. The teacher accepts the acknowledgement.
8. The administrator approves the course.
9. The system updates the course status and makes it available for enrollment or purchase.

Alternative flows

- The first lesson is incomplete → approval is withheld.
- The teacher does not accept the required acknowledgement → the state of the paid course will be pending and the administrator will be notified.
- The course violates platform policies → the administrator rejects it or requests changes.
- The administrator needs additional information → the course remains pending.

Postcondition: The course is published only after the required approval conditions have been satisfied.

UC-21 — Manage complaints

Actor: Administrator; Teacher or Teaching Assistant may handle complaints within their authorized scope.

Main flow

1. The administrator / teacher/ teaching assistant opens the complaints dashboard.
2. The system displays new and existing complaints.
3. The administrator / teacher/ teaching assistant opens a complaint and reviews its details.
4. The administrator or other authorized handler investigates the issue.
5. The administrator contacts the teacher when necessary.
6. The administrator/ teacher/ teaching assistant records the decision and any required action.
7. The system updates the complaint status.
8. Relevant parties are notified when required.

Alternative flows

- The complaint requires further investigation → its status remains Under Review and the writer is notified.
- The administrator needs a response from the teacher → the complaint moves to Waiting for Teacher.
- The complaint is resolved → the status becomes Resolved.
- The complaint is rejected → the status becomes Rejected and the decision is recorded.

Postcondition: The complaint has a traceable status, decision, and outcome.

UC-22 — Moderate course reviews

Actor: Administrator | **Request source:** Teacher or platform reports

Main flow

1. The administrator opens reported reviews or reviews requiring moderation.
2. The administrator inspects the review and its context.
3. The administrator checks whether the review violates the platform's moderation policy.
4. The administrator removes the review if it contains prohibited insults, abusive content, or other violations.
5. The system records the moderation action.

Alternative flows

- The review does not violate policy → it remains visible.
- The teacher disputes a review → the administrator evaluates the submitted request.
- A review is identified as spam or a duplicate → the administrator applies the defined moderation policy.

Business rule: Teachers may request review moderation, but the administrator is responsible for removal decisions under the current SRS.

UC-23 — Review refund requests

Actor: Administrator

Main flow

1. The administrator opens the refund requests queue.
2. The administrator selects a request.
3. The system displays the purchase details, refund deadline, and student's reason.
4. The administrator investigates the case.
5. The administrator approves or rejects the request.
6. If approved, the refund is processed through the supported financial procedure.
7. The system records the transaction and updates the relevant financial balances.
8. The student receives the decision.

Alternative flows

- The request is outside the applicable refund period → eligibility is evaluated according to policy.
- The purchase record cannot be verified → the request remains pending investigation.
- The refund fails → the financial case remains unresolved until reconciliation.

Postcondition: The request and financial outcome are traceable.

UC-26 — Manage course lifecycle

Actor: Administrator.

Main flow

1. The system identifies a course whose term or lifecycle has ended.
2. The course is archived after the term ends, according to the current decision.
3. The responsible teacher decides whether to retain or delete the archived course, subject to storage limits and platform policy.
4. The system applies the approved retention or deletion action.
5. The system handles related records according to the final data-retention policy.
6. The course ratings and reviews affect the teacher’s overall rating according to the policy.

Alternative flows

- The course has not reached the applicable lifecycle milestone → no lifecycle action occurs.
- The course is eligible for archival but the teacher has not decided → it remains archived pending a decision.
- The course is selected for permanent deletion → the system follows the approved deletion procedure.

9.8 Cross-Functional Scenarios

Scenario A — Paid Course Purchase and Teacher Earnings

Actors: Student, Platform, Payment Provider, Teacher

1. The student selects an approved paid course.
2. The system initiates payment and waits for the verified payment result.
3. If payment succeeds, the system creates the enrollment, grants immediate access, places the teacher's funds on hold, and starts the applicable refund period.
4. After the refund period, eligible earnings are released according to the payout policy.
5. If payment fails, the system does not confirm the purchase or grant paid access; it displays the payment status and allows a safe retry.

Scenario B — Student Learning and Grading

Actors: Student, Teacher, Teaching Assistant

1. The student opens an enrolled course and selects any module.
2. The student studies an available lesson and submits an assignment or takes a quiz when available.
3. The system saves the submission and updates progress.
4. The teacher or assigned teaching assistant grades work that requires manual grading.
5. The student views the result when it is available.

Key rule: Assignment grading does not block the student's next available learning step. Manual grading may remain pending while the student continues.

Scenario C — Paid Course Approval

Actors: Teacher, Administrator

1. The teacher creates a paid course and submits it for approval.
2. The administrator reviews course readiness and the required initial course content.
3. The teacher accepts the compliance acknowledgement.
4. If all approval conditions are satisfied, the administrator approves the course and it becomes available for purchase.
5. Otherwise, the course remains pending or is rejected until the issue is resolved.

9.9 Edge-Case Scenarios for Validation

Authentication and permissions

- A student tries to open the teacher dashboard.
- A teaching assistant tries to access a course they were not assigned to.
- A guardian attempts to view an unlinked student's grades.
- A logged-out user attempts to access protected course content.

Course access and assessments

- A student starts with Module 3 instead of Module 1.
- A student attempts Lesson 2 before completing Lesson 1.
- A student attempts a quiz without watching the lecture where access rules allow it.
- A student exceeds the permitted number of quiz attempts.
- A student submits an assignment and continues before it is graded.
- A teacher attempts to publish a module without its mandatory quiz.

Payments and financial integrity

- A payment fails or remains pending.
- The payment provider sends a duplicate transaction confirmation.
- A student requests a refund before the applicable deadline.
- A refund is approved but the transaction fails.
- A teacher views earnings while purchases remain within their refund periods.

Moderation and administration

- A teacher submits a request to remove an abusive review.
- A student submits a duplicate or spam review.
- A complaint is waiting for the teacher's response.
- A paid course is submitted without the required initial course content.

- A course reaches its archival or deletion milestone.

Reliability and data handling

- The network fails during quiz submission.
- An assignment submission fails before confirmation.
- A grade changes after manual review.
- A course is deleted while historical academic records exist.

10. Future Plan for the Project

The following features are planned for future versions and are outside the current core scope unless separately approved as a change to the SRS.

10.1 Live Code Editor

The platform may provide an integrated code editor where students can write code, select a programming language, run code, view output, and solve programming tasks.

10.2 Programming Assignments

Future versions may allow teachers to create programming problems. Students will be able to open a problem, write and run a solution, submit it, and receive teacher-assigned grades and feedback after review.

10.3 General Programming Tracks

The platform may expand beyond the school curriculum to offer additional educational tracks, including:

- Web Development.
- Mobile Development.
- General Programming Skills.
- Other programming technologies.